Dashboard

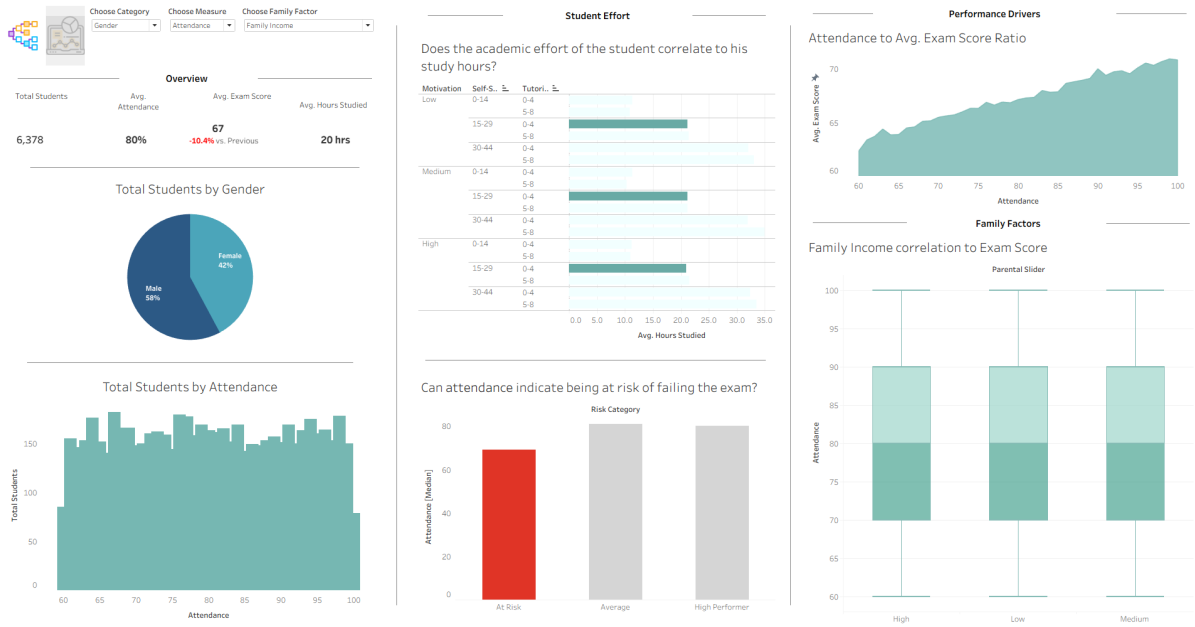

Figure 1: Analytics Dashboard Section

Figure 2: Data Classes Section

i. Main Audience and Justification

The intended audience of this dashboard includes academic Success coordinators and Department Heads. These groups have been identified as the intended audience of this dashboard because they are responsible for the well-being of individual students as well as the academic strategy of the university at large. The Success Coordinator uses the AE Dash tab to make tactical

interventions by identifying student behavioral risks such as low sleep or study hours. On the other hand, Department Heads use the "Clustered Dash" to understand trends and assess the impact of environmental factors such as university accessibility and peer pressure on student performance.

ii. **List of Charts and Justification for Inclusion**

The dashboard includes several specialized visualizations designed to provide a comprehensive look at student performance while maintaining a high data to ink ratio.

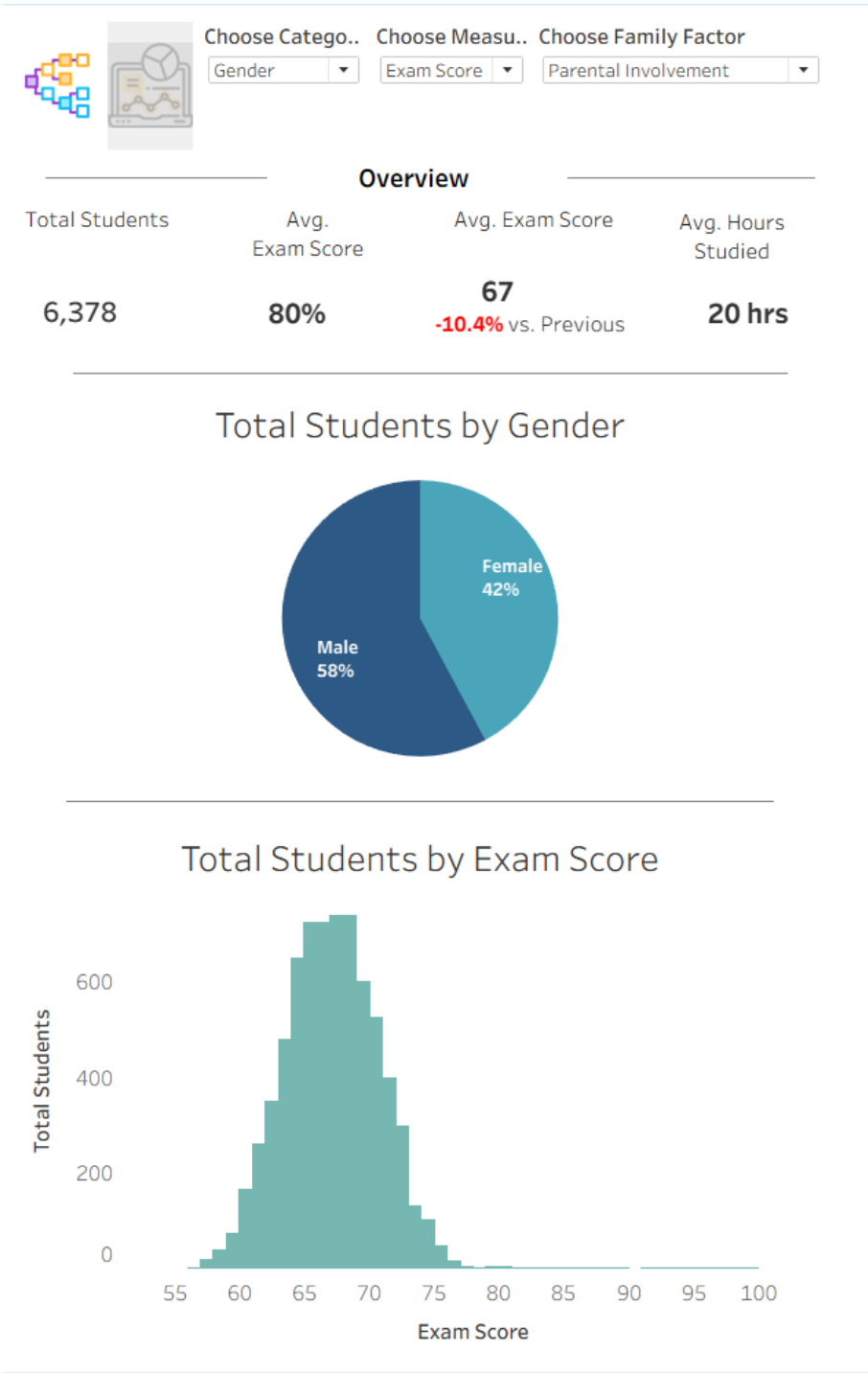

Figure 3: Overview KPIs and demographic distribution sidebar.

1. At the top of the behavioral effort view, I placed four large indicators that provide a high level snapshot of the student population. These include the total number of students, average attendance, average exam score, and average hours studied. A key feature here is the conditional formatting on the exam score, which shows a ten point four percent decline compared to previous results. This serves as the primary alert for the user.

2. Below the indicators, a dynamic sidebar allows users to pivot the entire view. It features a pie chart for categorical data and a histogram for numerical distribution. By using the choose category and choose measure parameters, users can toggle the pie chart to show income, peer influence, or school type, while the histogram shows the distribution of study or sleep hours.

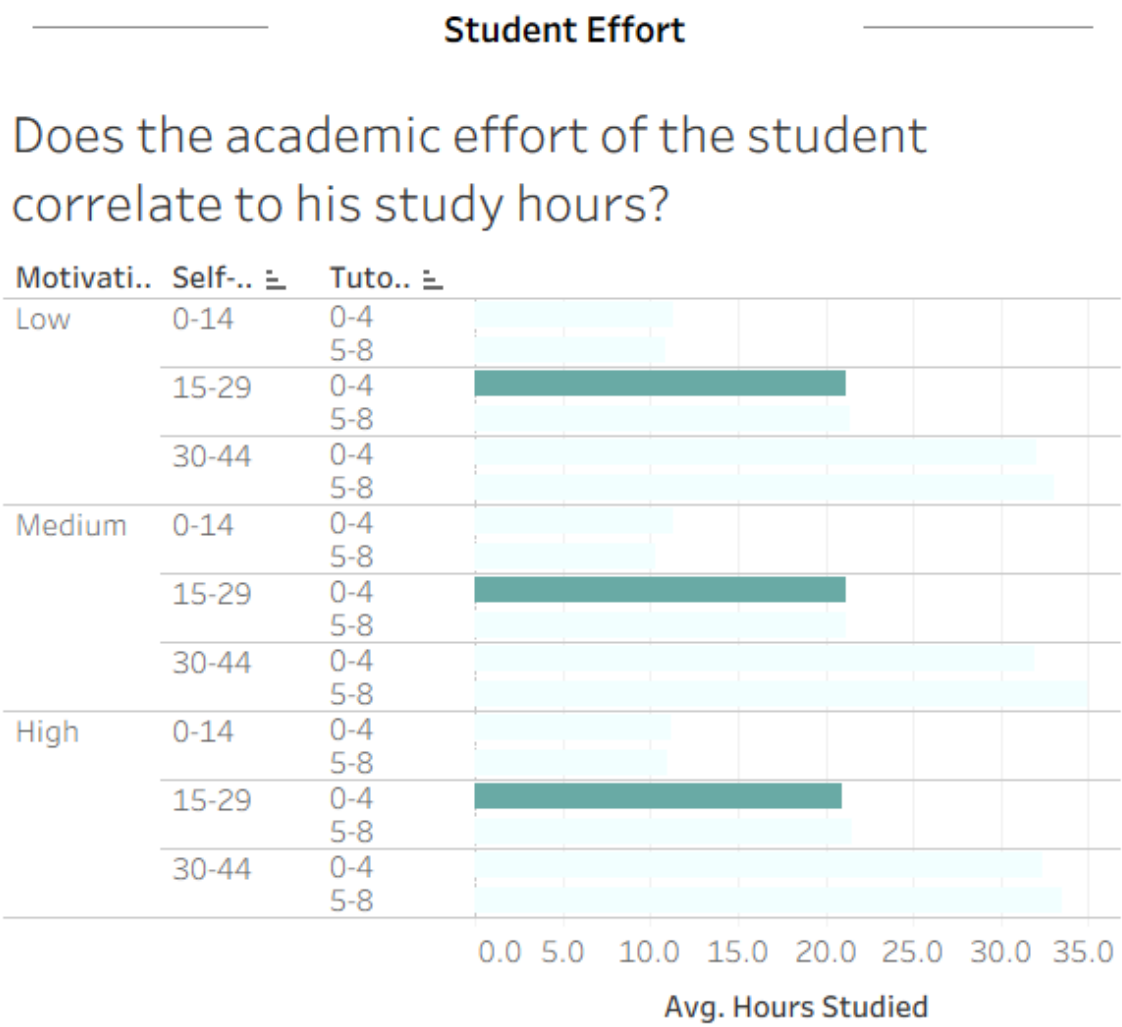

Figure 4: Breakdown of study hours by motivation and tutoring levels.

3. This grouped bar chart analyzes how motivation, self study, and tutoring sessions contribute to the total study hours of the student. It helps identify if high effort is driven by internal motivation or external support structures.

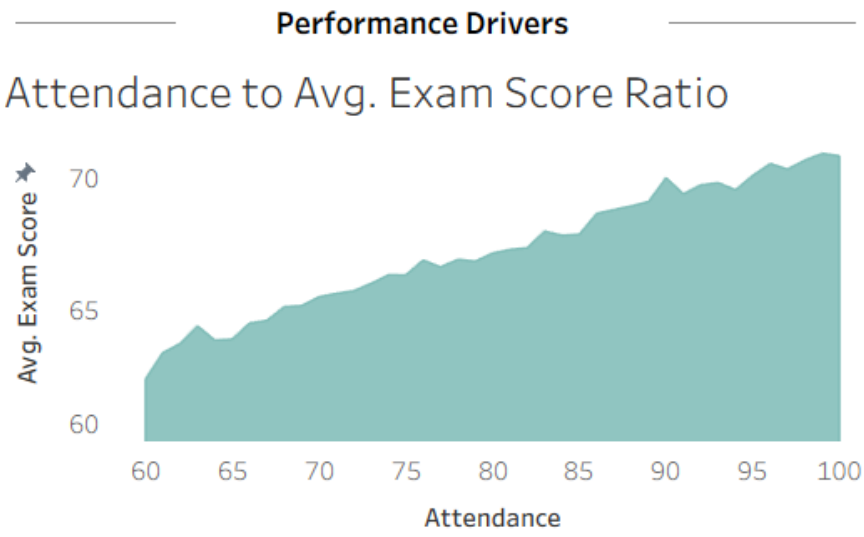

Figure 5: Interactive Measure-to-Measure Area Chart.

4. The performance drivers area chart is a fully interactive visual. Much like the risk chart, I have built this so that the user can switch between different measures. An advisor can use this to see the ratio of attendance to exam scores or sleep hours to exam scores. This justifies its inclusion by allowing the user to visually confirm the strength of different correlations across the student body.

5. The risk category chart is a summarized alert system that focuses on student outcomes. I have built this visual so that it is fully interactive. The user can use a parameter to switch the measure being viewed. This means an advisor can compare the at risk group against the high performers not only by exam scores but also by study hours or attendance. This flexibility is essential for seeing if an at risk student is struggling because of poor results or simply a lack of effort.

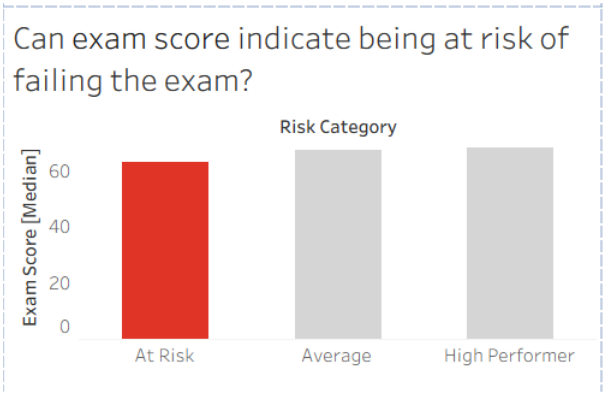

Figure 6: Interactive risk bar chart with measure switching capability.

6. The second dashboard view contains four high depth quadrants to analyze the root causes of performance variance. These are organized into social, university, lifestyle, and learning settings. Crucially, I have added a measure selector to this view as well. This allows a department head

to switch the entire environmental analysis between attendance, previous scores, or current exam results. This justifies the inclusion of these quadrants because they allow for a deep drill down into exactly which environmental barrier is most responsible for a drop in any given metric.

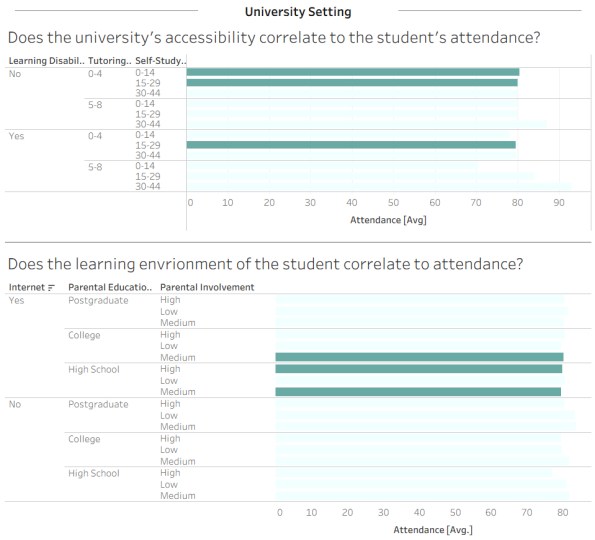

Figure 7: Two quadrants analyzing university and learning environments with dynamic measure switching.

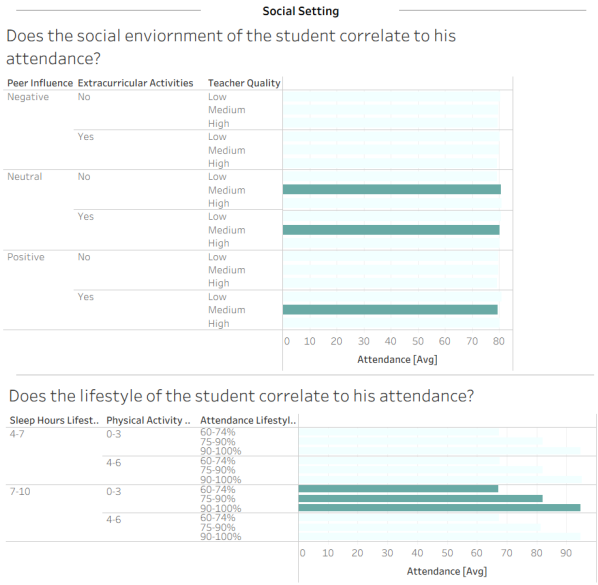

Figure 8: Two quadrants analyzing lifestyle and social with dynamic measure switching.

7. The family factors box and whisker plot is included to show the spread and variance of student scores. It is uniquely powerful because it allows the user to switch between three specific parameters: family income, parental involvement, and parental education level. This chart was included to show that while averages are helpful, the distribution and outliers within these socioeconomic groups reveal the true range of student success.

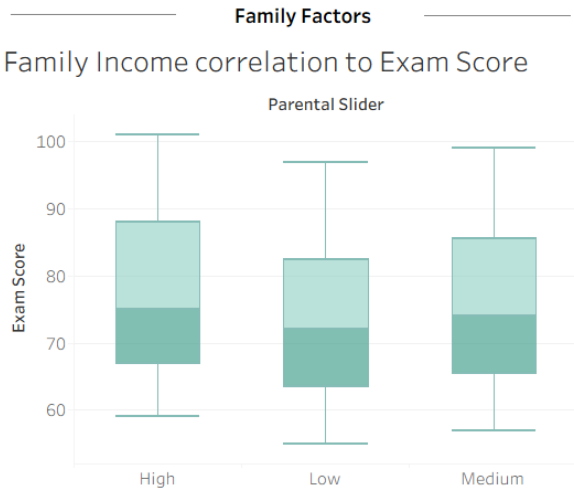

Figure 9: Socioeconomic distribution of scores using box and whisker analysis.

iii. **Deriving Key Insights from the Dashboard**

Through a number of interactive layers integrated into the system, the audience can gain insights. First, users can switch between behavioral effort and environmental clusters thanks to the dynamic dashboard switching. This aids an advisor in figuring out whether a student's failure is the result of institutional obstacles like university accessibility or personal habits.

Second, in-depth analysis of the performance data is made possible by the interactive parameters. The at-risk group frequently has longer sleep durations than average but much fewer study sessions than average, as can be found by using the measure selector on the risk chart. This knowledge enables a success manager to give a student more targeted time management guidance as opposed to merely general study advice.

Third, a comprehensive understanding of student life is made possible by the clustered environmental view's ability to switch measures. In the social setting quadrant, for instance, a coordinator can observe that when teacher quality is high, students who are negatively influenced by their peers still maintain high attendance. The user can then determine whether attendance truly results in success by changing the measure to exam scores.

The dashboard guides the user through this process with vertical separators and a unified color scheme. Gridlines are eliminated to keep the attention on the red risk alerts and the teal performance bars. For a department head who might only have a few minutes to review the data, this makes the insights simple to understand at a glance. Regardless of the type of school, consistent family support offers a more stable academic baseline. The parental slider further enables the user to see how parental involvement shifts the median score in the box plot.